



Note on Leverage Ratios

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Ratios used in financial management

- (i) Measuring overall performance:
 - Growth (e.g. in revenues)
 - Bottom line profitability (e.g. NI/Sales, ROE)
 - Stock-market performance (stock return, valuation multiples)
- (ii) Measuring operating (“real”) performance
 - Pricing (Gross margin, COGS/Sales)
 - Efficiency and scale (e.g. Op.exp./sales)
 - Operating margin (e.g. EBIT/Sales)
 - Asset efficiency (e.g. Assets / Sales)
- (iii) Measuring liquidity and working capital
 - Short-term liquidity (e.g. Current & Quick ratios)
 - Working capital efficiency (e.g. NWC/Sales, days inventory/receivables/payables)
- (iv) **Measuring capital structure**
 - The focus of this hand-out

Ratios: generally

- Ratios can only be interpreted in a **relative** sense
 - By looking at changes over time / trends
 - By comparing with other competitors & other industries
- When interpreting ratios, relate to economics of business
 - Can we understand historical numbers / trends given the firm's past?
 - Do ratios make sense given industry economics (growth, competition)?
- Many possible ratios could measure any given phenomenon
 - E.g. can normalize profits with sales or assets, can use end of year balance sheet values or BOY/EOY averages.
 - » Use common sense given industry and what you are trying to capture.
 - Common with industry-specific ratios that capture relevant value-drivers / KPIs

Measuring leverage

- I use three main ratios: “stock”, “flow”, and “mixed”
 - Commonly used by lenders and credit rating agency’s for measuring company indebtedness.
- 1. “Stock” measure: **Debt to capital**
- Net debt to total capital = $(D - C) / (D + E - C)$,
 - D = interest bearing debt
 - C = liquid assets, including cash and short-term investments.
 - E = book equity, I.e. net worth
 - Motivation?

■ Why interest bearing debt rather than total liabilities?

- Only want to capture liabilities due to financing, not liabilities that all-equity firms would also have (def. taxes, A/P, accruals, etc.)
 - » We will cover working capital financing issues in the P&G-case
- Makes sense to also include capitalized value of future lease payments in D, since this is similar to debt.
 - » Note: We will typically not have data on PV of leases and other off-balance sheet liabilities in the cases we cover in this course.

■ Why net out cash?

- Cash = negative debt.
 - » Could use cash to pay off debt instantly.
- In principle, **only excess cash should be deducted**
 - » $\text{Excess cash} = (\text{total cash}) - (\text{operating cash})$
 - » Operating cash = cash needed to operate efficiently
 - » We will discuss how to estimate operating cash in the valuation cases week 2 (e.g. Roche & Genentech)
- As long as comparable firms face similar operating cash needs, netting out total vs excess cash will typically not affect conclusions in any major way.

■ Why book value of equity and not market value?

- Goal of capital structure analysis is to make sure that the company can avoid distress in a bad case scenario.
Market cap today >> market cap in bad case scenario.
 - » Value of firm in bad case closer to value of book assets.
- A major determinant of debt capacity is the collateral value of the firm's assets. This is likely to be closer to book value than to market value.
- Credit analysts and debt rating agencies generally care more about book than market leverage.

■ However: when *unlevering betas* to calculate cost of capital, use *market* leverage ratios.

- Why? Because the CAPM theory says so.

2. “Flow” measure: **interest coverage ratios**.

- Commonly used: EBITDA / I or EBIT / I , where
 - I = net interest expense = interest expense – interest income.
- Very natural measure of leverage: **does firm earn enough money to make debt payments and avoid default.**
- If have more information on debt structure, can do better:
 - Instead of I , can use total debt service (TDS), where $\text{TDS} = \text{net interest expense} + \text{principal repayments}$ (approx. by interest bearing debt due in a year)
 - Instead of EBIT or EBITDA, normalize with EBITA (deducting depreciation but not amortization of intangible assets) or with Free Cash Flow (FCF).

3. “Mixed” measure: **Debt/EBITDA**

- (Approximately) answers question: “How many years will it take to pay off debt?”
 - » As before, can consider net debt: $(\text{Debt} - \text{Cash}) / \text{EBITDA}$
- Debt/EBITDA standard leverage measure in highly levered transactions, such as LBOs
 - » When leverage level is assumed not to be permanent / optimal for the long-term
- Alternatively: Debt/EBIT, Debt/EBITA, or Debt / $(\text{FCF} - (1-t)*I)$
 - » How many years does it take to pay down debt *without* having to cut necessary CAPEX?